

Week 2 - Rotations

Recap: Qubits, Logic Gates

Quantum Computers manipulate delicate physical states in a computationally useful way.

A qubit is the fundamental unit of quantum computation:

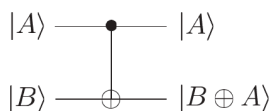
1. It can take the values $|0\rangle, |1\rangle$, but also an arbitrary linear combination of them:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

2. Dirac notation is the convention in QC: $|_ \rangle$ is called a ket, it signifies a quantum state, just like 0 and 1 are classical states
3. α and β are complex numbers. $|\alpha| = \sqrt{\alpha\alpha^*}$. On measuring the qubit, $|\alpha|^2$ is the probability of measuring the qubit in the 0 state, and $|\beta|^2$ for the 1 state. To conserve probability, $|\alpha|^2 + |\beta|^2 = 1$
4. A bitstring of length n is a series of bits like 000 or 0101. Bitstrings of length n take 2^n values from 0...0 to 1...1
5. A register (group) of n qubits spans a 2^n -dimensional space, with basis states being the 2^n bitstrings: One qubit's basis is $(|0\rangle, |1\rangle)$, two qubit's basis is $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$.
6. An arbitrary state of n qubits is a superposition (linear combination) of this basis:
 $a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle$, where $|a_{00}|^2 + |a_{01}|^2 + |a_{10}|^2 + |a_{11}|^2 = 1$
7. So we can simply represent the state of n qubits by a column vector of 2^n complex coefficients: $[a_{00}, a_{01}, a_{10}, a_{11}]^T$, where I write it as the transpose of a row vector for convenience.
8. Logic gates which act on these vectors are simply $2^n \times 2^n$ complex matrices.
9. The only constraint on such a matrix for it to be a logic gate is that it should be **unitary**:
 $UU^\dagger = I$, where the dagger means take the complex conjugate, then transpose.
10. We call unitary matrices, unitaries for short.
11. This course doesn't worry about how these matrices arise (see last year's Learner's Space for that). Rather we will understand how these matrices are broken down into simpler gates, so that they can be implemented on hardware.

Recap: Quantum Circuits

controlled-NOT



$$U_{CN} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

1. A Quantum Circuit is a sequence of unitaries which act on n qubits. The circuit diagram has n wires (one per qubit) with the input register on the left, data flowing from left to right.
2. If the diagram shows a unitary only acts on a subset of wires, it acts as identity on the rests.
3. We evaluate a circuit on a given register, by going left to right, and premultiplying the unitary on the vector

Representing these objects in code

We will work in Python, and the numpy library provides complex vectors and matrices.

For example:

```
import numpy as np

# Use a single complex dtype for numpy everywhere.
```

```
DTYPE=np.complex128

state = np.array([1, 0], dtype=DTYPE)

INV_SQRT2 = 1.0 / np.sqrt(2.0)

# Hadamard Gate:
H = INV_SQRT2 * np.array([[1, 1], [1, -1]], dtype=DTYPE)

print(H @ state) # gives [0.70710678+0.j 0.70710678+0.j]
```

Rotations in 3D

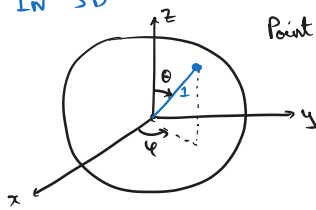
2D RECAP



Rotation matrix: $R(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$ $R(\alpha) \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} = \begin{bmatrix} \cos(\theta + \alpha) \\ \sin(\theta + \alpha) \end{bmatrix}$
 by Trigo identities
 $\det(R_2) = 1$

2D Rotation descriptions $\rightarrow$ angle θ
 $\rightarrow$ Rotation matrix $\begin{bmatrix} \sim & \sim \\ \sim & \sim \end{bmatrix}$

In 3D



Point $(u, \theta) = \begin{bmatrix} \cos \theta \sin \theta \\ \sin \theta \sin \theta \\ \sin \theta \end{bmatrix}$

3D Rotation descriptions $\rightarrow$ Rotation Matrix: eg $R_x(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$
 $\rightarrow$ Euler Angle: $(\hat{n}, \theta)$ unit vector $\downarrow$
 every rotation is a rotation about some axis $\hat{n}$ by angle θ

$M^\dagger$ ^{adjoint}
 $=$ conjugate transpose

Unitary: $UU^\dagger = U^\dagger U = I$

Hermitian: $M = M^\dagger$

THE PAULI MATRICES

$X = \sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$Y = \sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$Z = \sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = -i\sigma_1\sigma_2\sigma_3 = I$

$\sigma_x\sigma_y = i\sigma_z$

$\nearrow$ cycle
 $\sigma_y\sigma_z = -i\sigma_x$

eg $\sigma_z\sigma_y = -i\sigma_x$

Other important unitaries:

Hadamard $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Phase $S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$\pi/8$ $T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$

FACT: Any Hermitian unitary with trace=0 can be written as:
 $M = M^\dagger = M^{-1}$, $\text{tr}(M) = 0$:

$$M = n_1 \sigma_1 + n_2 \sigma_2 + n_3 \sigma_3 \xrightarrow{\text{written as dot product}} \hat{n} \cdot \vec{\sigma} \longrightarrow \text{Pauli vector } \vec{\sigma} = (\sigma_1, \sigma_2, \sigma_3)$$

where $\hat{n}$ is a unit real vector

$$\text{i.e. } n_1^2 + n_2^2 + n_3^2 = 1, \quad n_1, n_2, n_3 \in \mathbb{R}$$

For an angle ϕ , $\phi \hat{n}$ is the Euler angle representation for rotation about $\hat{n}$ by ϕ of the Bloch vector

Then unitary which does that rotation is

$$R_{\hat{n}}(\phi) = e^{-i\phi \hat{n} \cdot \vec{\sigma} / 2} = \cos \phi I + i \sin \phi (\hat{n} \cdot \vec{\sigma})$$

exponential of a matrix: see MATH10

← This is another representation of 3D rotations: 2x2 unitaries

FACT: Any 2x2 unitary can be written as

$$U = e^{i\alpha} e^{i\phi \hat{n} \cdot \vec{\sigma}}$$

$\downarrow$ Global phase $\searrow$ some rotation

$$\left[\begin{array}{l} \alpha = \frac{1}{2} \arg(\det U) \longrightarrow \text{Let } \tilde{U} = e^{-i\alpha} U \\ \cos \frac{\theta}{2} = \frac{1}{2} \text{tr}(\tilde{U}) \quad n_j \sin \frac{\theta}{2} = \frac{i}{2} \text{tr}(\sigma_j \tilde{U}) \end{array} \right.$$

Git repo for coding tasks

This Git repository <https://github.com/jjbel/quantum-circuits-LS> is a template. Clone the repo to your own account, and fill in the functions. Later, you will have to submit the link to your repo.

Task-1: Find the Bloch Representation of a 2x2 Unitary

In `rotation.py`, fill in the `to_bloch` function which finds the phase, axis, and angle of a single-qubit gate using the formulae above.

Universal Basis for Single-Qubit Gates

Quantum Computer hardware only supports a discrete finite set of single-qubit gates, called the universal basis. For example H and T (defined above).

This week, let's tackle decomposing 2x2 unitaries into a universal basis.

Every unitary has a Bloch rotation representation. Euler's rotation theorem states that any 3D rotation can be broken into a sequence of 3 rotations about 2 orthogonal axes:

e.g. $U = R_X(\alpha)R_Y(\beta)R_X(\gamma)$, where X and Y are the orthogonal axes, XYX is the sequence, and we can find the angles α, β, γ for any U . (ignoring the global phase)

Sadly $R_X(\alpha)$ is a family of infinite matrices, so we can't use it in a universal basis. Given a single rotation matrix like H , we can apply it repeatedly H^n repeats the rotation of H , n times.

Unfortunately most basic gates like X, Y, Z, T, H rotate by an angle ϕ which is a rational multiple of π , so their powers only span a discrete finite set of angles: e.g. $H^2 = I$ so its powers either rotate by 0° or 180°

Instead, consider the unitaries M_1, M_2 defined below. They rotate by an angle which is an irrational multiple of π , about two orthogonal axes n_1, n_2 . Thus we can decompose any U as $R_{n1}(\alpha)R_{n2}(\beta)R_{n1}(\gamma)$. Then we can **approximate** $R_{n1}(\alpha)$ using powers of M_1 :

Given an acceptable error e , we can find k such that $k\lambda\pi \bmod 2\pi$ can come within e of α . Similarly for β, γ

Thus, U is decomposed as $M_1^k M_2^l M_3^m$

Thus M_1 and M_2 form a universal basis.

In fact, M_1 and M_2 can in turn be written as products of powers of the more basic matrices H and T .

Thus H, T form a universal basis which can approximate any unitary arbitrarily accurately.

$$\boxed{T^8 = I = Z^2} \quad \text{UNIVERSAL SET}$$

$$Z^{1/4} = T \quad Z^{-1/4} = T^\dagger$$

$$X^{1/4} = H Z^{1/4} H \\ = H T H$$

$$\text{Let } M_1 = e^{i\lambda\pi\hat{n}_1 \cdot \vec{\sigma}} = Z^{-1/4} X^{1/4} T^\dagger H T H$$

$$\begin{aligned} \text{Let } M_2 &= e^{i\lambda\pi\hat{n}_2 \cdot \vec{\sigma}} = H^{-1/2} Z^{-1/4} X^{1/4} H^{1/2} \\ &= H^{-1/2} T^\dagger H T H H^{1/2} \\ &= \underbrace{T^2 H T H T^6 H T^\dagger H T^6}_{H^{-1/2}} \underbrace{T^\dagger H T H}_{M_1} \underbrace{T^2 H T H T^2 H T^\dagger H T^6}_{H^{1/2}} \end{aligned}$$

$$\text{Then } \cos \lambda\pi = \cos^2 \frac{\pi}{8} = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right) \rightarrow \text{This means } \lambda \text{ is irrational}$$

$$\begin{aligned} \hat{n}_1 &= \sqrt{2} \cos \frac{\pi}{8} \frac{\hat{z} - \hat{x}}{\sqrt{2}} + \hat{y} \\ \hat{n}_2 &= \sqrt{2} \cos \frac{\pi}{8} \hat{y} - \frac{\hat{z} - \hat{x}}{\sqrt{2}} \end{aligned} \quad \leftarrow \text{Remember to normalize these}$$

Task 2: Rotation decomposition into orthogonal axes

Given a Bloch object with angle ϕ about axis n , find its α, β, γ angles in the n_1, n_2, n_3 decomposition using:

$$\cos \phi = \cos \beta \cos(\gamma + \alpha)$$

$$\hat{n} \sin \phi = \hat{n}_1 \cos \beta \sin(\gamma + \alpha)$$

$$+ \hat{n}_2 \sin \beta \cos(\gamma - \alpha)$$

$$+ \hat{n}_3 \times \hat{n}_2 \sin \beta \sin(\gamma - \alpha)$$

Fill this in the function `n1n2n3_angles()`

Task 3: Approximate angles with powers

Fill in `approx_angle_with_tolerance()` to approximate an arbitrary angle as a multiple of $\lambda\pi$ (defined above).

Task 4: Full decomposition

Put the previous tasks together: fill in `decompose_2x2()`

It accepts a unitary, converts it to Bloch form, finds the $n_1 n_2 n_1$ angles, then approximates them.

Week 1 Doubts

Double Cover: why the half-angle in Bloch representation

The set of 3x3 real rotation matrices is called $SO(3)$. The set of 2x2 unitaries with determinant 1 (because we remove the global phase) is called $SU(2)$.

Notice that given a member of $SU(2)$, U then $-U$ also has determinant 1 because of dimension 2.

$-U$ also represents the same rotation (try to work this out by hand!)

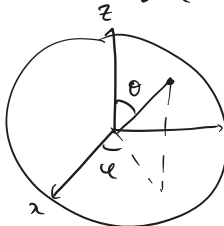
Thus $SU(2)$ covers $SO(3)$ twice: for a given 3D rotation, there are two possible unitaries which represent it. Look up the belt trick and quaternions, to read more about this.

This is reflected in the half-angle in the rotation matrix formula, and in the Bloch sphere representation.

See also: <https://quantumcomputing.stackexchange.com/a/4121>

$\langle \psi | Z | \psi \rangle$ is the expectation value of the Z-coordinate of the Bloch vector (treat this as a fact).

Start with $|0\rangle : \theta=0, \varphi=90^\circ (\pi/2)$
 consider $\begin{bmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos \frac{\theta}{2} \\ i \sin \frac{\theta}{2} \end{bmatrix} \equiv \frac{\theta}{2} = \frac{\varphi}{2}$



$\varphi \in [0, 2\pi)$
 $\theta \in [0, \pi)$

$$|\psi\rangle = \begin{bmatrix} \cos \frac{\theta}{2} \\ e^{i\varphi} \sin \frac{\theta}{2} \end{bmatrix}$$

$$\langle \psi | := (|\psi\rangle)^\dagger = \begin{bmatrix} \cos \frac{\theta}{2} & e^{-i\varphi} \sin \frac{\theta}{2} \end{bmatrix}$$

$$\langle \psi | \sigma_z | \psi \rangle = \cos \frac{\theta}{2} - \sin \frac{\theta}{2} = \cos \theta$$

Two-Level Gates are not Two-Qubit Gates

n qubits have 2^n basis vectors, and are acted upon by a $2^n \times 2^n$ unitary.

A two-level gate acts as identity on all but a 2-dimensional subspace spanned by 2 bases (these are the 2 "levels"). For example:

$$\tilde{U} = \begin{pmatrix} a & & & & & b \\ & 1 & & & & \\ & & 1 & & & \\ & & & 1 & & \\ & & & & 1 & \\ c & & & & & d \end{pmatrix}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in U(2).$$

This is analogous to how a rotation matrix around the x-axis is

$$: \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}$$

It only rotates in the y-z plane (these are the two levels) leaving x untouched.

A two-*qubit* gate would have a full 4×4 unitary acting on a 4-dimensional subspace.
For a single qubit, there only are two-levels, and unitaries are 2×2 .